

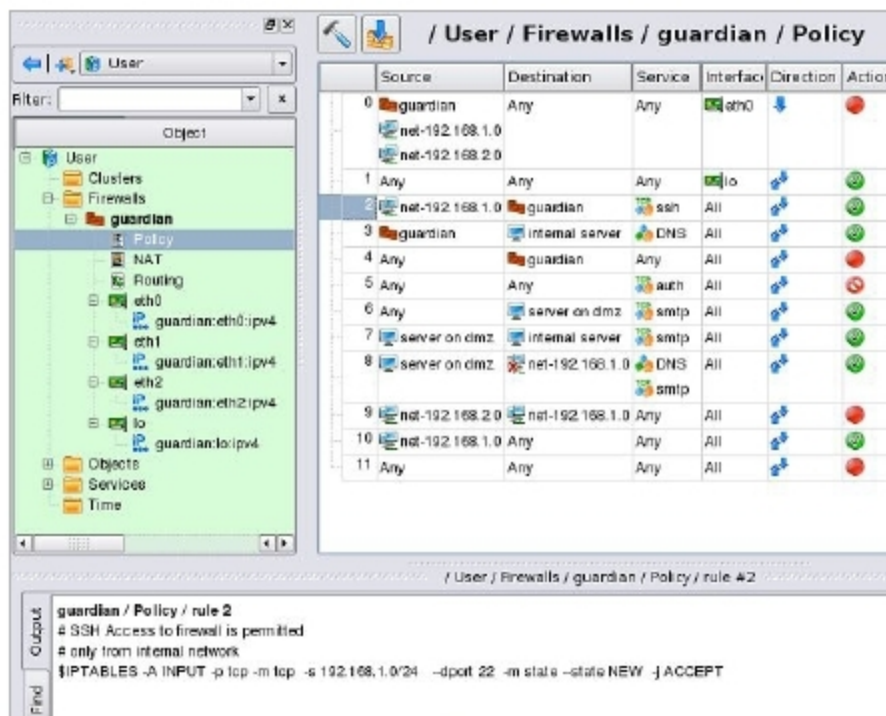


Figure 1: Firewall Builder GUI

just generalisations of the ideas and features found in all of those firewalls. The program helps you create and manage rule sets and then translates them into the configuration language of the chosen target firewall platform.

Firewall Builder is more complex than many basic firewall configuration GUIs such as Firestarter, but on the other hand, one can build very complex policies with it and fully use the flexibility and power of iptables and other supported firewalls. Firewall Builder works well both as a configuration tool for the local firewall protecting machine where fwbuilder is running and as a firewall configuration UI that manages configuration on several remote dedicated firewalls and routers.

Fwbuilder allows for the simultaneous management of multiple firewalls on many servers. Programming components called policy compilers facilitate fine tuning of firewall policy rules without the need for the administrator to remember all of the details for each firewall or server. The policy compilers also ensure that firewalls are ‘sane’—that there are no contradictions or errors that will interfere with the intended and legitimate use of the network by its subscribers.

How Firewall Builder works

Firewall Builder GUI provides a simple and easy interface to build a customised firewall. The systems administrator can create and manage collections of objects that define network addresses, hosts, firewalls, services and design policies as well as NAT rules using these objects.

The process involves the following five steps.

1. **Creating the firewall:** Firewall Builder lets the user define all sorts of settings like the type of platform, software version and interfaces. When a user clicks on the ‘Create New Firewall’ button, a wizard based interface follows to let the user configure the following:
 - The firewall’s name

- Software on which the firewall is based (like Cisco Pix or iptables)
 - Operating system on which the firewall will execute
 - Select from pre-configured firewall templates
 - Select Ethernet interfaces, etc.
2. **Defining objects:** Firewall Builder enables admins to create objects for network elements, for use in firewall rules. It comes preconfigured with hundreds of commonly used objects. Firewall Builder stores objects in what are known as libraries. By default, it includes the following two libraries.
 - **Standard:** The predefined objects that can be dragged and dropped into firewalls. These objects cannot be edited.
 - **User:** An empty pre-categorised library to which users can add their own objects and then drag and drop them into firewalls. The following are the types of objects in Firewall Builder.
 - **Address ranges:** A range of addresses can be configured into a single object.
 - **Address tables:** This is an address-based object that can be created when a range of addresses is needed, but the actual addresses are not known when the firewall or policy is being written. The address tables object has an added feature that allows for the object to be loaded at either compile time (during firewall compilation) or during run time (when Firewall Builder runs the firewall script).
 - **Address:** A single address that can be used for an interface, source or destination (such as a host).
 - **DNS names:** This object represents a DNS ‘A’ or ‘AAAA’ name, and resolves to an IP address during either compile or run time.
 - **Groups:** A group is a container that holds references to multiple objects of the same or similar type (addresses, address ranges, network objects, etc).
 - **Hosts:** A host object represents hosts on a network — desktops, workstations and any other network node that has a network address.
 - **Networks:** This object describes an IP network or an entire subnet.
- To create a new object, the following steps are required:
- Right-click the *Hosts* entry in the left navigation menu (under *objects*).
 - Select ‘New Host’.
 - Give the new host a name and click *Next*.
 - Select *Configure interfaces manually* and click *Next*.
 - Enter the information for the host to be configured (name, label, static IP, IP address) and click *Finish*.
- This allows admins to create an unlimited number of hosts. The hosts are added to the object tree, and it is possible to drag and drop them into the firewall.
3. **Configure policy:** Firewall Builder makes use of objects defined to specify the rules of the firewall. To create a new policy on the host, expand the navigation tree for *Hosts*, find the desired host to be added, and drag and drop the host into the firewall.